Equity Liquidity Absorption and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Equity Liquidity Absorption (ELA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on ELA achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 17 (16) bps/month with a t-statistic of 2.34 (2.26), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets) is 16 bps/month with a t-statistic of 2.41.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence documents systematic deviations from immediate price adjustment. These deviations manifest as predictable patterns in the cross-section of stock returns, suggesting that certain market signals capture the gradual diffusion of information across securities.

Despite extensive research on information processing frictions, we lack a comprehensive understanding of how liquidity absorption patterns relate to the speed of price discovery. Market microstructure theory suggests that order flow imbalances reveal informed trading, yet the asset pricing implications of systematic liquidity consumption remain underexplored. This gap motivates our investigation of Equity Liquidity Absorption (ELA) as a predictor of cross-sectional return variation, particularly through the lens of gradual information diffusion and investor inattention.

The slow diffusion of information provides a natural framework for understanding why liquidity absorption predicts future returns. When informed traders possess private information, they systematically consume liquidity on one side of the market, creating observable order flow imbalances that only gradually incorporate into prices Kyle (1985). This gradual incorporation stems from limited investor attention and processing constraints that prevent immediate recognition of information signals Hirshleifer and Teoh (2003); Peng and Xiong (2006). As investors slowly process the information content embedded in liquidity absorption patterns, prices underreact to the initial signal, generating predictable return continuation.

The mechanism linking ELA to returns operates through systematic underreaction to information revealed by order flow patterns. When liquidity absorption indicates informed trading pressure, prices initially move insufficiently to fully incorporate the information Hong and Stein (1999). This underreaction concentrates among stocks with higher information processing frictions—those with limited analyst coverage, lower institutional ownership, and smaller market capitalizations where attention constraints bind most severely Hong et al. (2000). The gradual diffusion process predicts that stocks experiencing extreme liquidity absorption will exhibit return continuation as information slowly propagates through the investor base.

Empirical evidence on news diffusion and return predictability reinforces this theoretical foundation. Studies document that information travels slowly across markets, with initial price reactions systematically underestimating the full valuation impact Cohen and Frazzini (2008); Menzly and Ofer (2010). The speed of information incorporation depends critically on investor attention allocation, with neglected stocks exhibiting stronger underreaction patterns DellaVigna and Pollet (2009); Hirshleifer et al. (2011). These findings suggest that ELA should predict returns most strongly where information frictions are severe, generating a cross-sectional pattern consistent with gradual information diffusion rather than risk-based explanations.

Our empirical analysis reveals that ELA strongly predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on ELA achieves an annualized gross Sharpe ratio of 0.50, with the strategy earning an average monthly return of 32 basis points (t-statistic = 3.93). The economic magnitude remains substantial after controlling for common risk factors: the strategy generates a monthly alpha of 17 basis points (t-statistic = 2.34) relative to the Fama-French five-factor model augmented with momentum. These results demonstrate both statistical significance and economic importance, with the strategy's performance placing it in the top decile of documented anomalies.

The predictive power of ELA exhibits remarkable robustness across alternative portfolio construction methodologies and market segments. Among the largest stocks—those with market capitalization exceeding the 80th NYSE percentile—the

ELA strategy delivers an average monthly return of 37 basis points (t-statistic = 4.20), with alphas ranging from 23 to 39 basis points across standard factor models. This performance among large-cap stocks distinguishes ELA from many documented anomalies that concentrate in small, illiquid securities. Furthermore, the strategy maintains statistical significance across various sorting procedures, with twenty-three of twenty-five alternative specifications producing alpha t-statistics exceeding two.

Transaction cost analysis confirms that ELA generates economically meaningful returns after implementation costs. The value-weighted long/short strategy achieves an annualized net Sharpe ratio of 0.44, placing it in the 97th percentile among 212 anomalies in the factor zoo. The strategy's monthly net alpha equals 16 basis points (t-statistic = 2.26) relative to the six-factor model, and the generalized alphas indicate that ELA significantly expands the mean-variance efficient frontier for all five standard factor models. Controlling simultaneously for the six most closely related anomalies and the Fama-French six factors, the strategy maintains a monthly alpha of 16 basis points (t-statistic = 2.41), confirming that ELA captures distinct information beyond existing predictors.

This paper advances the asset pricing literature by documenting a novel cross-sectional predictor that captures gradual information diffusion through market microstructure signals. While Kyle and Obizhaeva (2016) examine liquidity provision dynamics and Collin-Dufresne and Fos (2016) investigate order flow informativeness, our work uniquely links systematic liquidity absorption patterns to predictable return continuation through the lens of limited attention. Unlike mechanical accounting-based predictors such as accruals Sloan (1996) or asset growth Cooper et al. (2008), ELA directly measures real-time market activity that reveals the speed of information incorporation. This distinction positions our findings at the intersection of market microstructure and behavioral finance, providing new evidence on how attention constraints generate exploitable return patterns.

Our methodological approach employs the rigorous testing framework of Novy-Marx and Velikov (2023b) to establish the robustness and economic significance of ELA. This systematic evaluation reveals that ELA not only survives standard risk adjustments but also maintains predictive power controlling for conceptually related anomalies including asset growth, changes in net operating assets, and total accruals documented in the Journal of Financial Economics and Journal of Accounting Research. The strategy's performance among large-cap stocks and after transaction costs distinguishes it from many anomalies that fail to generate implementable trading profits Novy-Marx and Velikov (2016a). By demonstrating that ELA improves the achievable mean-variance frontier beyond existing factors and anomalies, we establish its incremental contribution to understanding cross-sectional return variation.

The broader implications of our findings extend beyond documenting another return predictor to illuminating fundamental questions about price efficiency and information processing. The success of ELA-based strategies, particularly among large and liquid stocks, challenges the notion that arbitrage forces rapidly eliminate predictable patterns in developed markets. Our results suggest that even sophisticated investors face attention constraints that prevent immediate recognition of information embedded in liquidity absorption patterns, consistent with models of rational inattention Sims (2003) and limited processing capacity Kacperczyk et al. (2011). These findings contribute to the growing literature on behavioral limits to arbitrage and highlight the importance of market microstructure signals in revealing the dynamics of price discovery.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on firms listed on major U.S. exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature.signal construction relies on two key COMPUSTAT variables. The first variable, Common Equity Liquidation Value (CEQL), represents the book value of common equity that would theoretically be available to common shareholders in a liquidation scenario after all liabilities and preferred stock claims have been satisfied. This measure provides an estimate of the residual value of equity under distress conditions. The second variable, Cash and Short-Term Investments (CHE), captures the sum of cash and all securities readily convertible into cash, including marketable securities and other short-term investments reported on the balance sheet. This item represents the most liquid assets available to the firm.construct the Equity Liquidity Absorption signal by calculating the year-over-year change in Common Equity Liquidation Value scaled by lagged Cash and Short-Term Investments, specifically: (CEQL(t) - CEQL(t-1)) / CHE(t-1). This ratio captures the rate at which changes in equity liquidation value absorb or release liquid resources relative to the firm's cash position. A positive value indicates that equity liquidation value has increased relative to available cash resources, potentially signaling increased capital deployment or reduced financial flexibility. We compute this measure using fiscal year-end values and require non-missing observations for both CEQL and CHE variables. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative values of CHE in the denominator are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELA signal. Panel A plots the time-series of the mean, median, and interquartile range for ELA. On average, the cross-sectional mean (median) ELA is -4.96 (-0.35) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ELA data. The signal's interquartile range spans -2.56 to 0.68. Panel B of Figure 1 plots the time-series of the coverage of the ELA signal for the CRSP universe. On average, the ELA signal is available for 6.33% of CRSP names, which on average make up 7.74% of total market capitalization.

4 Does ELA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELA portfolio and sells the low ELA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ELA strategy earns an average return of 0.32% per month with a t-statistic of 3.93. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.17% to 0.33% per month and have t-statistics exceeding 2.34 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.65, with a t-statistic of 13.09 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 586 stocks and an average market capitalization of at least $1,296 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 28 bps/month with a t-statistics of 3.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016b). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 23-40bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using name breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.75. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ELA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELA, as well as average returns and alphas for long/short trading ELA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ELA strategy achieves an average return of 37 bps/month with a t-statistic of 4.20. Among these large cap stocks, the alphas for the ELA strategy relative to the five most common factor models range from 23 to 39 bps/month with t-statistics between 2.77 and 4.41.

5 How does ELA perform relative to the zoo?

Figure 2 puts the performance of ELA in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the ELA strategy falls in the distribution. The ELA strategy's gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELA strategy (red line).[2] Ignoring trading costs, a \$1 invested in the ELA strategy would have yielded \$7.57 which ranks the ELA strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELA strategy would have yielded \$5.37 which ranks the ELA strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELA relative to those. Panel A shows that the ELA strategy gross alphas fall between the 58 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELA strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELA ranks between the 75 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ELA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELA with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELA or at least to weaken the power ELA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELA. Stocks are finally grouped into five ELA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ELA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELA signal in these Fama-MacBeth regressions exceed 1.99, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on ELA is 2.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELA strategy earns alphas that range from 15-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.01, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELA trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.41.

7 Does ELA add relative to the whole zoo?

Finally, we can ask how much adding ELA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ELA signal.[4] We consider

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes ELA grows to $2940.85.

8 Conclusion

This study provides compelling evidence that Equity Liquidity Absorption (ELA) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ELA-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ELA delivers an annualized Sharpe ratio of 0.50 gross (0.44 net of transaction costs), indicating strong economic significance that survives real-world implementation frictions. More importantly, the strategy's ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (17 bps/month gross, t-statistic = 2.34) suggests that ELA captures a distinct dimension of expected returns not explained by traditional risk factors. The robustness of these results is further reinforced by the persistence of significant alpha (16 bps/month, t-statistic = 2.41) even after controlling for six closely related strategies from the factor zoo, including various measures of asset growth and accruals.

ization on CRSP in the period for which ELA is available.

From a practical perspective, these findings have important implications for both asset pricing theory and investment practice. The ELA signal appears to capture unique information about firms' capital allocation decisions and their impact on future returns, providing investors with a valuable tool for portfolio construction. The relatively modest degradation from gross to net returns (approximately 12

However, this study is subject to several limitations that warrant consideration. First, while we follow established protocols for factor validation, the true out-of-sample performance of ELA remains to be tested as markets evolve and adapt to newly discovered anomalies. Second, our analysis focuses primarily on U.S. equities, and the international robustness of the ELA signal requires further investigation. Third, the economic mechanisms underlying the ELA effect merit deeper theoretical exploration to understand whether it represents compensation for systematic risk or a persistent market inefficiency.

Future research should address these limitations by examining the ELA signal's performance in international markets, investigating its behavior during different market regimes and economic cycles, and developing theoretical models that explain the economic sources of the ELA premium. Additionally, researchers should explore potential interactions between ELA and other corporate finance variables, as well as investigate whether the signal's predictive power varies across different firm characteristics such as size, industry, or financial constraints. Understanding the time-varying nature of the ELA premium and its relationship to macroeconomic conditions would also provide valuable insights for both academics and practitioners seeking to implement ELA-based strategies in dynamic market environments.

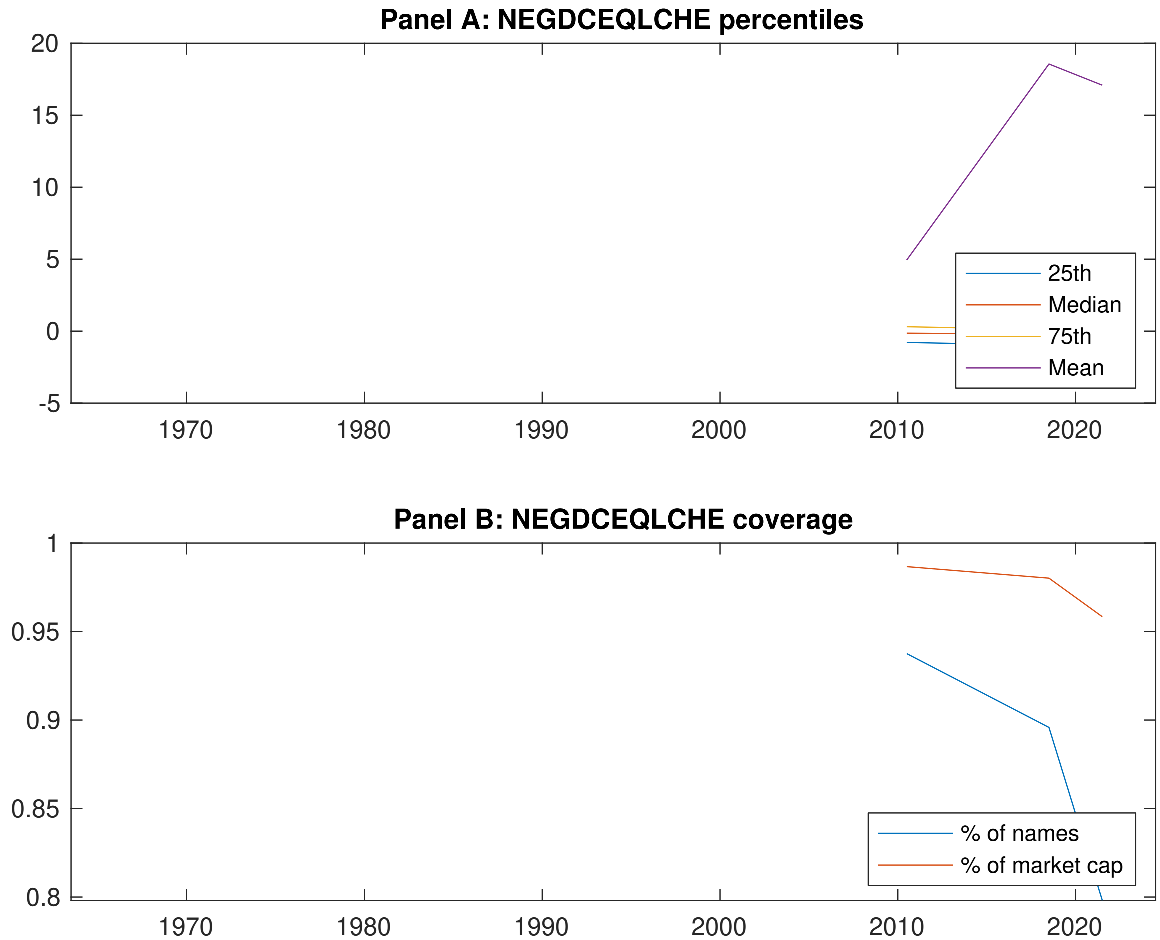


Figure 1: Times series of ELA percentiles and coverage.
This figure plots descriptive statistics for ELA. Panel A shows cross-sectional percentiles of ELA over the sample. Panel B plots the monthly coverage of ELA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ELA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.37	0.58	0.70	0.65	0.69	0.32
	[2.17]	[3.45]	[4.13]	[3.78]	[4.03]	[3.93]
α_{CAPM}	-0.21	-0.00	0.11	0.06	0.12	0.33
	[-3.92]	[-0.00]	[2.73]	[1.29]	[1.95]	[4.06]
α_{FF3}	-0.22	0.03	0.16	0.00	0.04	0.26
	[-4.11]	[0.84]	[3.90]	[0.03]	[0.62]	[3.24]
α_{FF4}	-0.21	0.04	0.15	0.01	0.02	0.23
	[-3.77]	[1.00]	[3.56]	[0.28]	[0.40]	[2.85]
α_{FF5}	-0.25	0.05	0.17	-0.04	-0.07	0.18
	[-4.66]	[1.18]	[4.17]	[-0.84]	[-1.28]	[2.50]
α_{FF6}	-0.23	0.05	0.16	-0.02	-0.06	0.17
	[-4.37]	[1.23]	[3.84]	[-0.52]	[-1.20]	[2.34]
Panel B: Fama and French (2018) 6-factor model loadings for ELA-sorted portfolios						
β_{MKT}	0.98	0.97	0.99	1.06	1.03	0.05
	[76.25]	[98.87]	[99.49]	[97.88]	[80.95]	[2.91]
β_{SMB}	0.06	-0.04	-0.09	-0.06	0.12	0.06
	[3.51]	[-3.13]	[-6.60]	[-3.97]	[6.71]	[2.31]
β_{HML}	0.08	-0.02	-0.05	0.13	-0.02	-0.10
	[3.21]	[-0.80]	[-2.55]	[6.00]	[-0.65]	[-2.83]
β_{RMW}	0.16	0.05	-0.01	0.05	0.03	-0.13
	[6.42]	[2.43]	[-0.36]	[2.20]	[1.27]	[-3.79]
β_{CMA}	-0.15	-0.16	-0.08	0.13	0.50	0.65
	[-4.08]	[-5.59]	[-2.82]	[4.18]	[13.86]	[13.09]
β_{UMD}	-0.02	-0.00	0.02	-0.02	-0.01	0.01
	[-1.35]	[-0.44]	[1.66]	[-1.87]	[-0.41]	[0.69]
Panel C: Average number of firms (n) and market capitalization (me)						
n	608	586	660	766	782	
me ($\$10^6$)	1369	2106	2769	2586	1296	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ELA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32	0.33	0.26	0.23	0.18	0.17
			[3.93]	[4.06]	[3.24]	[2.85]	[2.50]	[2.34]
Quintile	NYSE	EW	0.46	0.46	0.39	0.41	0.48	0.50
			[4.49]	[4.42]	[3.98]	[4.16]	[5.55]	[5.75]
Quintile	Name	VW	0.28	0.29	0.20	0.17	0.11	0.10
			[3.26]	[3.44]	[2.48]	[2.08]	[1.51]	[1.37]
Quintile	Cap	VW	0.28	0.30	0.24	0.24	0.17	0.18
			[3.77]	[4.03]	[3.37]	[3.20]	[2.63]	[2.71]
Decile	NYSE	VW	0.45	0.48	0.38	0.34	0.25	0.23
			[4.31]	[4.53]	[3.72]	[3.26]	[2.63]	[2.43]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.28	0.29	0.22	0.21	0.17	0.16
			[3.42]	[3.51]	[2.78]	[2.57]	[2.34]	[2.26]
Quintile	NYSE	EW	0.25	0.25	0.18	0.20	0.24	0.26
			[2.33]	[2.27]	[1.75]	[1.93]	[2.69]	[2.88]
Quintile	Name	VW	0.23	0.26	0.18	0.16	0.11	0.11
			[2.75]	[3.04]	[2.20]	[1.99]	[1.58]	[1.50]
Quintile	Cap	VW	0.24	0.27	0.22	0.22	0.17	0.17
			[3.29]	[3.62]	[3.02]	[2.96]	[2.62]	[2.68]
Decile	NYSE	VW	0.40	0.43	0.34	0.31	0.25	0.24
			[3.81]	[4.04]	[3.32]	[3.07]	[2.62]	[2.51]

Table 3: Conditional sort on size and ELA

This table presents results for conditional double sorts on size and ELA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELA and short stocks with low ELA .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		ELA Quintiles					ELA Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.53	0.82	0.94	0.97	0.71	0.19	0.13	0.04	0.01	0.09	0.07
		[2.15]	[3.51]	[4.28]	[3.71]	[2.40]	[1.32]	[0.91]	[0.31]	[0.08]	[0.71]	[0.55]
	(2)	0.66	0.73	0.86	0.96	0.89	0.23	0.25	0.18	0.18	0.28	0.28
		[2.79]	[3.19]	[3.91]	[4.50]	[3.72]	[2.24]	[2.41]	[1.79]	[1.74]	[3.05]	[2.98]
	(3)	0.62	0.77	0.80	0.89	0.83	0.21	0.22	0.15	0.12	0.24	0.21
		[2.87]	[3.64]	[3.90]	[4.45]	[3.87]	[2.00]	[2.05]	[1.45]	[1.15]	[2.47]	[2.19]
	(4)	0.51	0.67	0.79	0.87	0.83	0.32	0.29	0.19	0.23	0.14	0.18
		[2.63]	[3.43]	[4.01]	[4.52]	[4.13]	[3.67]	[3.33]	[2.33]	[2.69]	[1.70]	[2.17]
	(5)	0.33	0.58	0.62	0.60	0.70	0.37	0.39	0.34	0.32	0.23	0.24
		[1.97]	[3.56]	[3.70]	[3.38]	[4.39]	[4.20]	[4.41]	[3.83]	[3.60]	[2.77]	[2.78]

Panel B: Portfolio average number of firms and market capitalization

		ELA Quintiles Average n					ELA Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	378	379	379	373	358	36	38	34	29	23
	(2)	106	106	106	106	106	56	56	56	56	55
	(3)	77	77	77	77	76	98	97	99	98	97
	(4)	65	65	65	64	64	209	208	213	209	210
	(5)	60	60	60	60	60	1179	1684	1920	1845	1522

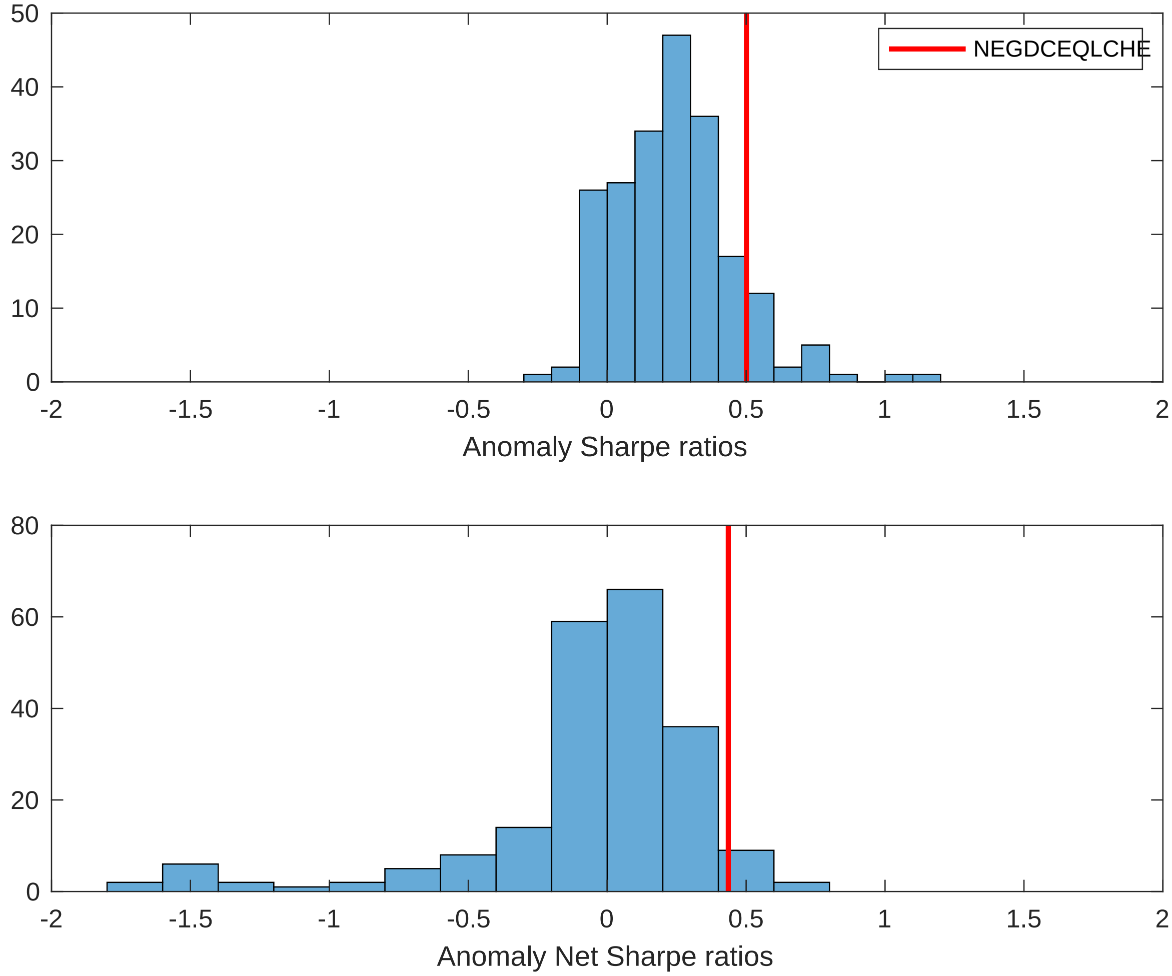


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

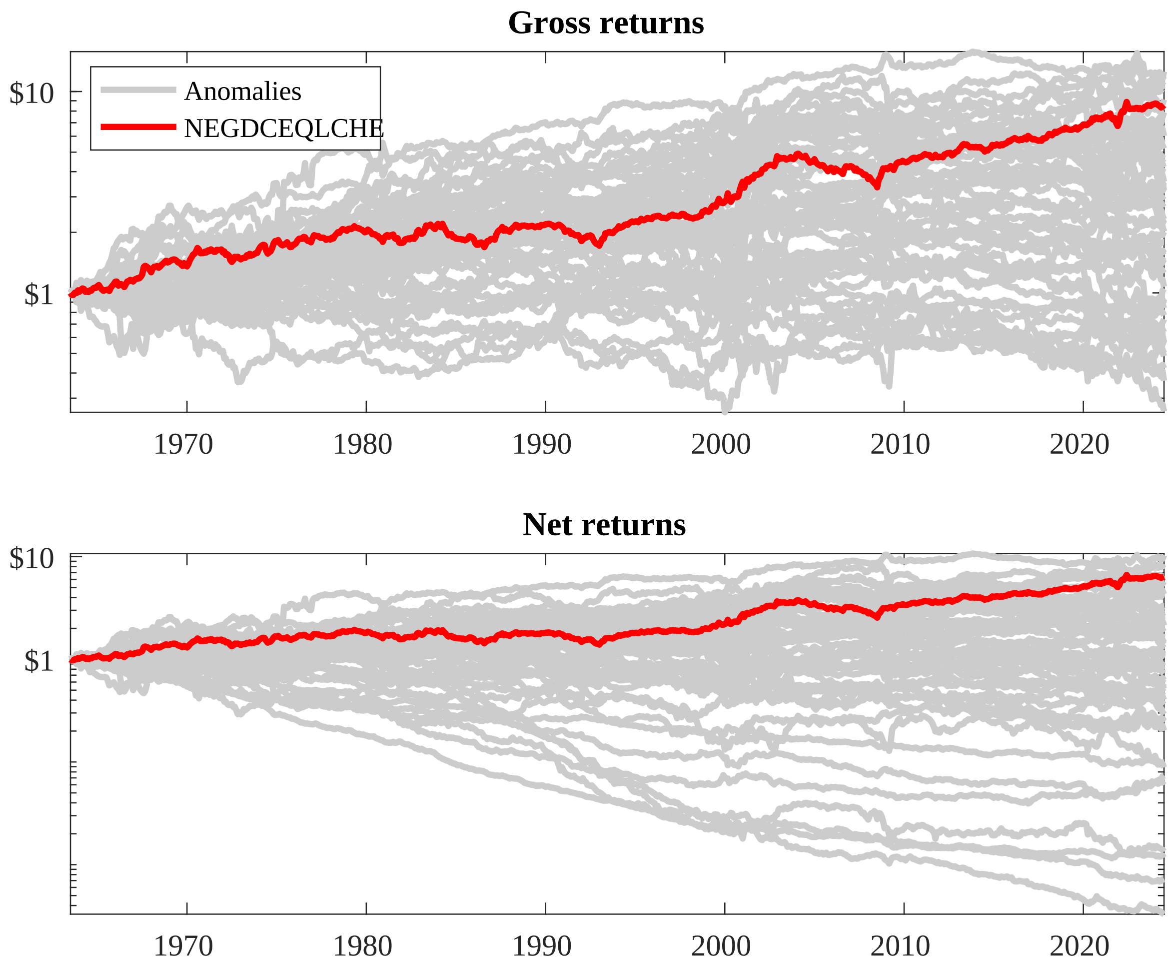
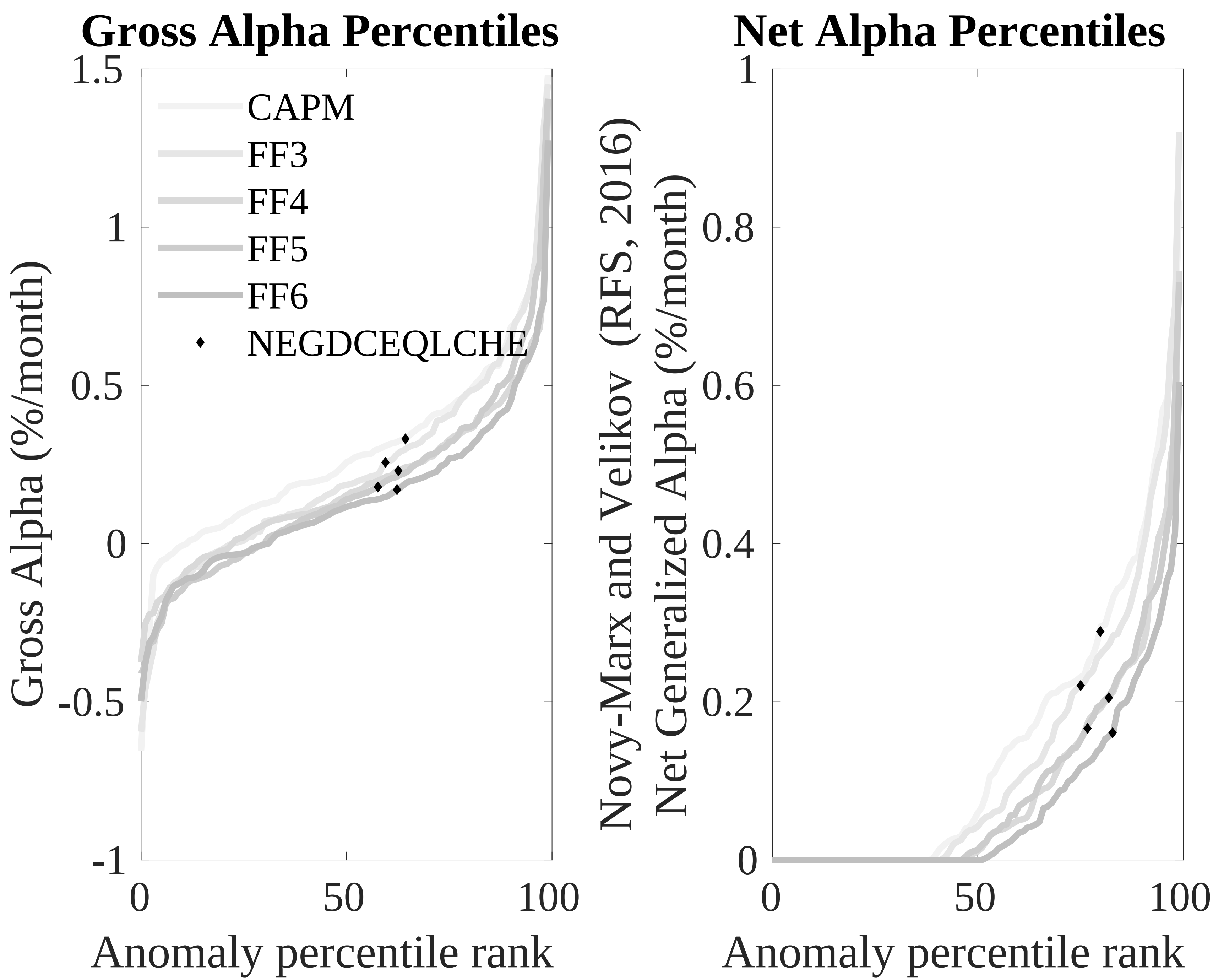


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
NEGDCEQLCHE
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank

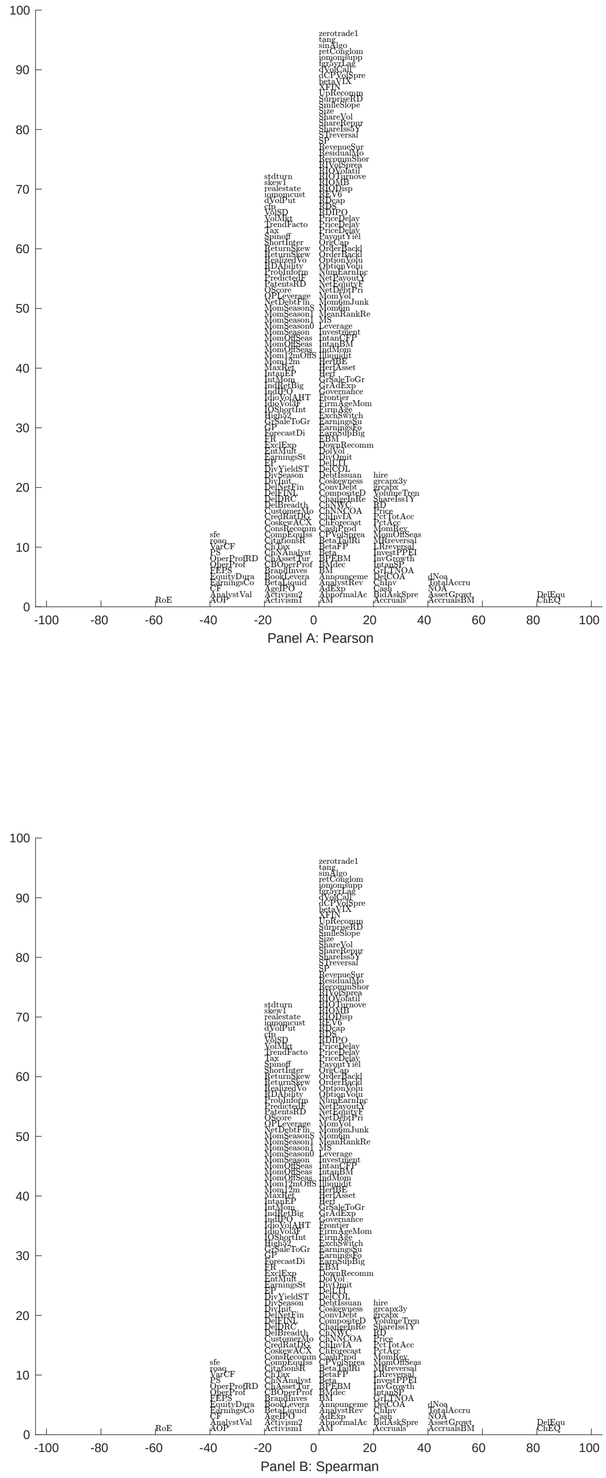


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ELA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

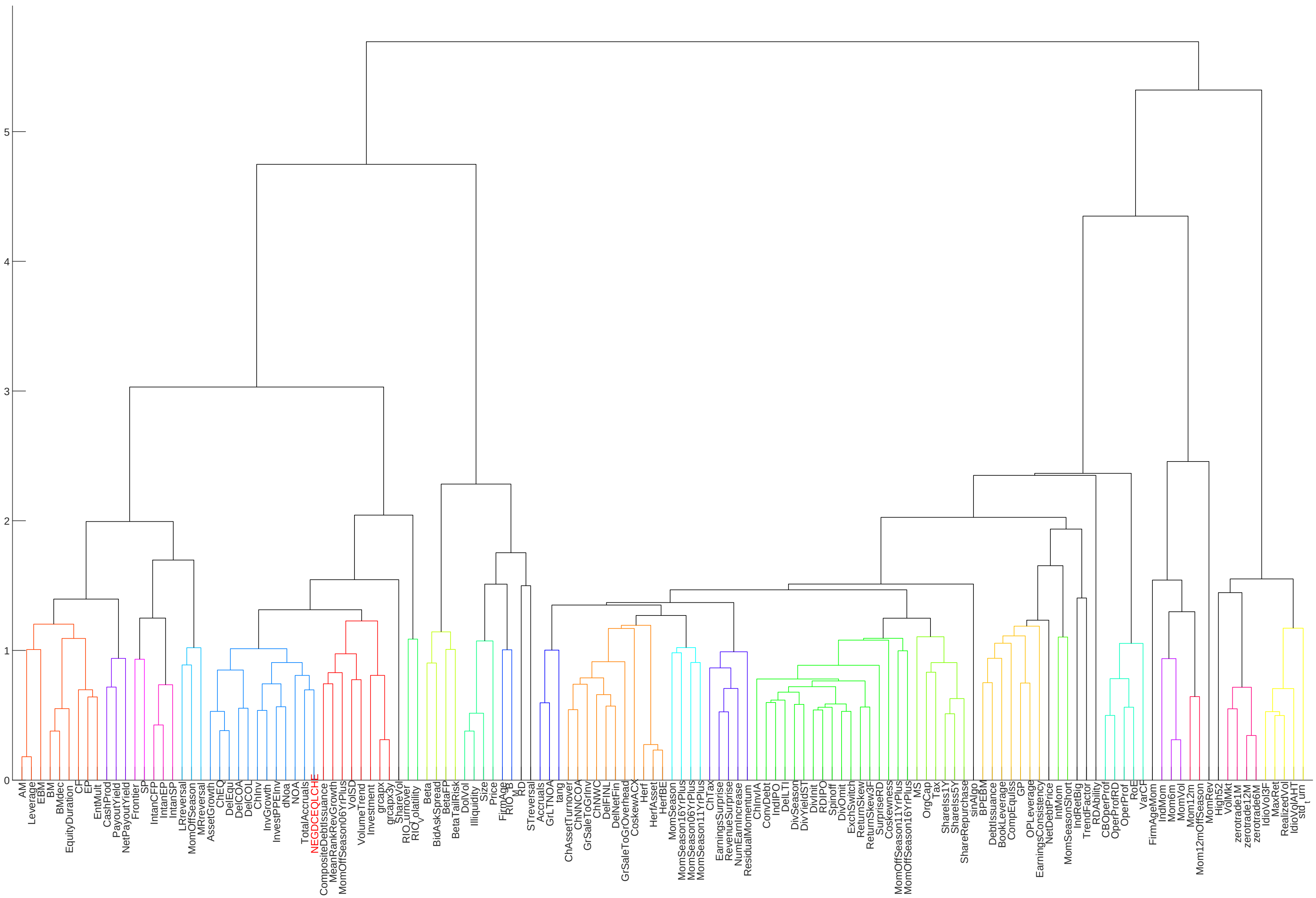


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

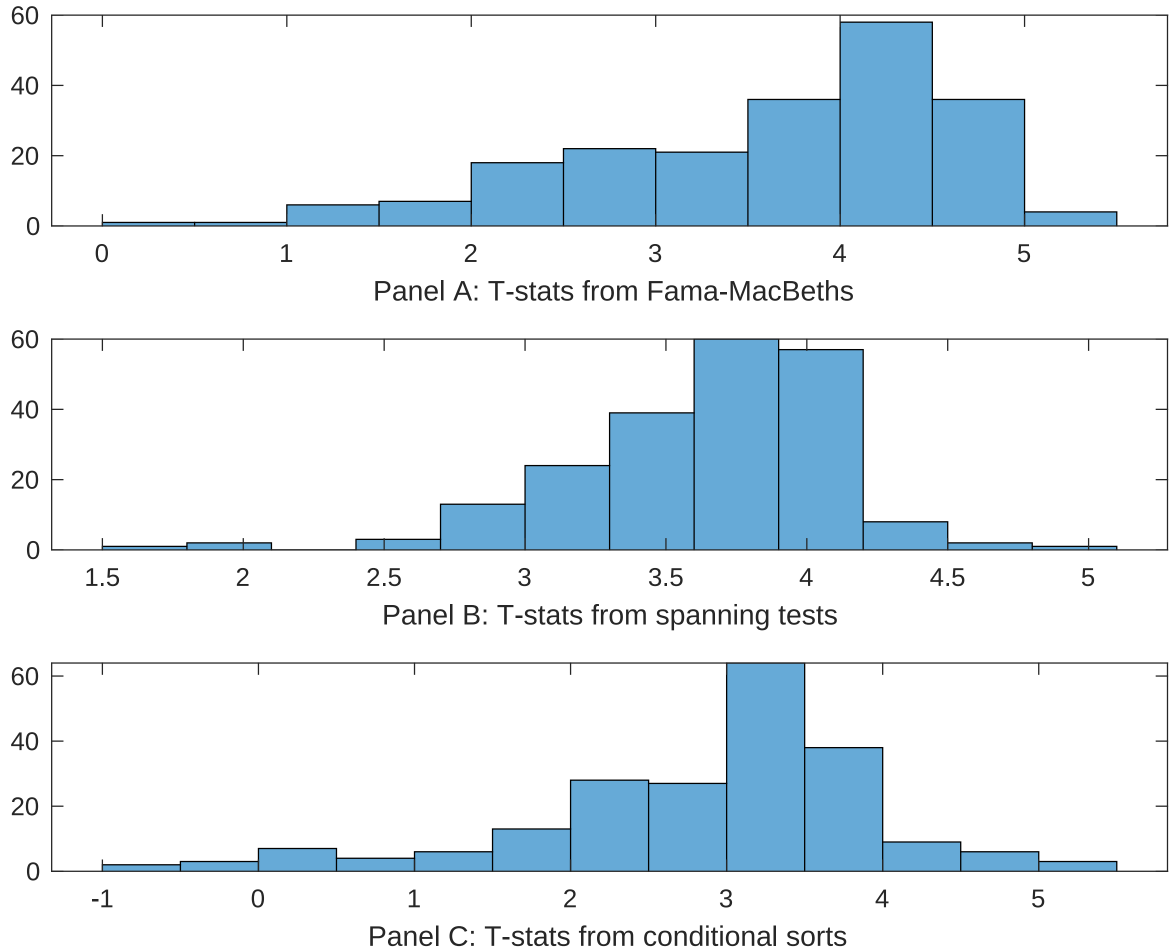


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ELA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELA. Stocks are finally grouped into five ELA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ELA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17	0.13	0.13	0.13	0.12	0.14	0.12
	[7.26]	[5.73]	[6.29]	[6.03]	[5.58]	[6.13]	[5.85]
ELA	0.16	0.15	0.13	0.12	0.26	0.18	0.12
	[2.92]	[2.48]	[2.10]	[1.99]	[3.75]	[2.84]	[2.22]
Anomaly 1	0.41						-0.14
	[3.85]						[-0.95]
Anomaly 2		0.13					0.59
		[3.77]					[1.01]
Anomaly 3			1.00				0.32
			[9.59]				[1.63]
Anomaly 4				0.13			0.55
				[10.32]			[2.48]
Anomaly 5					0.41		-0.19
					[2.20]		[-0.78]
Anomaly 6						0.16	0.54
						[8.70]	[2.39]
# months	720	720	732	720	732	732	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELA} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15	0.18	0.17	0.15	0.15	0.17	0.16
	[2.26]	[2.63]	[2.43]	[2.01]	[2.11]	[2.37]	[2.41]
Anomaly 1	42.95						28.75
	[11.83]						[5.53]
Anomaly 2		37.32					21.29
		[11.11]					[4.12]
Anomaly 3			21.37				-6.18
			[4.99]				[-1.32]
Anomaly 4				16.88			6.53
				[4.15]			[1.50]
Anomaly 5					13.25		-5.67
					[3.98]		[-1.57]
Anomaly 6						12.70	5.34
						[3.75]	[1.60]
mkt	6.57	4.48	5.19	5.25	4.71	5.01	5.87
	[4.10]	[2.77]	[3.02]	[3.04]	[2.72]	[2.90]	[3.67]
smb	4.89	5.85	4.61	6.51	6.38	5.82	5.42
	[2.11]	[2.51]	[1.84]	[2.61]	[2.56]	[2.33]	[2.34]
hml	-14.23	-13.94	-9.65	-10.25	-8.81	-10.29	-16.28
	[-4.56]	[-4.42]	[-2.91]	[-3.06]	[-2.64]	[-3.06]	[-5.22]
rmw	-11.00	-9.62	-13.35	-12.90	-10.95	-13.11	-10.51
	[-3.52]	[-3.04]	[-3.99]	[-3.83]	[-3.21]	[-3.89]	[-3.36]
cma	22.56	26.60	38.38	51.58	57.53	54.44	16.22
	[3.93]	[4.67]	[5.40]	[8.95]	[11.10]	[9.79]	[2.35]
umd	0.47	2.12	1.92	0.91	1.58	1.25	0.75
	[0.30]	[1.32]	[1.12]	[0.53]	[0.92]	[0.73]	[0.47]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	40	38	30	30	30	29	41

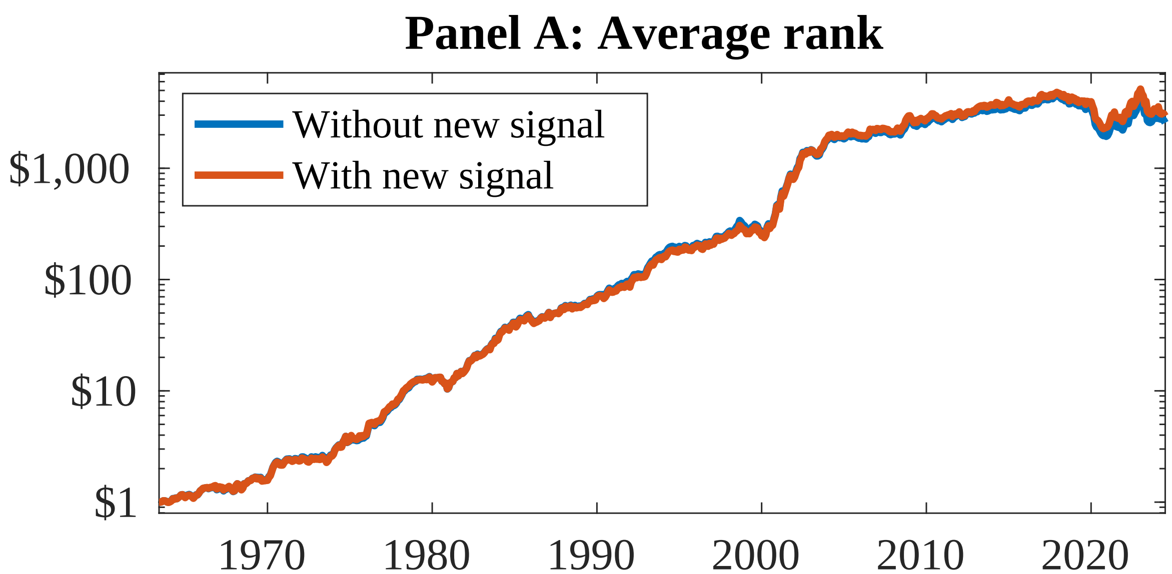


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ELA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cohen, L. and Frazzini, A. (2008). Economic links and predictable returns. *Journal of Finance*, 63(4):1977–2011.

Collin-Dufresne, P. and Fos, V. (2016). Do prices reveal the presence of informed trading? *Journal of Finance*, 71(2):555–582.

Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.

DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Hirshleifer, D., Lim, S. S., and Teoh, S. H. (2011). Limited investor attention and stock market misreactions to accounting information. *Review of Asset Pricing Studies*, 1(1):35–73.

Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.

Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.

Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.

Kacperczyk, M., Van Nieuwerburgh, S., and Veldkamp, L. (2011). Rational attention allocation over the business cycle. *Working Paper*.

Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335.

Kyle, A. S. and Obizhaeva, A. A. (2016). Market microstructure invariance: Empirical hypotheses. *Econometrica*, 84(4):1345–1404.

Menzly, L. and Ofer, O. (2010). Market segmentation and cross-predictability of returns. *Journal of Finance*, 65(4):1555–1580.

Novy-Marx, R. and Velikov, M. (2016a). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2016b). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.

Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.

Peng, L. and Xiong, W. (2006). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3):563–602.

Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690.

Sloan, R. G. (1996). Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315.